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Key words

Floating marine structures; Offshore renewable energy; Fluid-structure interactions; Shared mooring.

Work Experience

2024.09 – Now **Teaching Assistant (80 h per year), Research Assistant**, Universidad Politécnica de Madrid.

Education

- 2023.09 – Now **Ph.D. Universidad Politécnica de Madrid, Spain**
Naval architecture and ocean engineering
Supervisor: Associate Prof. Javier Calderon-Sanchez; Prof. Zhiyu Jiang
- 2026.08 – 2026.11 **Visiting Ph.D. Shanghai JiaoTong University, China**
Supervisor: Prof. Zhen Gao (Academician of the Norwegian Academy of Technological Sciences)
Research topic: Shared mooring, FPV.
- 2025.05 – 2025.08 **Visiting Ph.D. University of Agder, Norway**
Supervisor: Prof. Zhiyu Jiang (Academician of the Norwegian Academy of Technological Sciences)
Research topic: Floating PV, Anti-collision protection for OWT, Shared mooring.
- 2020.09 – 2023.06 **M.Eng. Wuhan University of Technology, China**
Hydraulic engineering.
Supervisor: Prof. Jin Pan; Prof. Xiaobin Li
- 2016.09 – 2020.07 **B.Eng. Wuhan University of Technology, China**
Harbor, waterway, and coastal engineering.

Selected Research Publications

13 published SCI journal articles, **8** of them as First/Corresponding Author (*), **1** Highly Cited Paper
Google Scholar Citation: 289; h-index=8; i10-index=6 (2026.07)

Journal Articles

- 13** H. Peng, X. Li, **W. Z. Zhang***, J. Pan, and P. Hu, “Physics-informed multi-layer perceptron (MLP) model with data augmentation for ship-bridge collision safety assessment,” **Ocean Engineering (Q1, IF 6.3)**, vol. 362, p. 126 339, 2026. 🔗 DOI: 10.1016/j.oceaneng.2026.126339.
- 12** **W. Z. Zhang***, Z. Jiang, J. Calderon-Sanchez, *et al.*, “Wind-assisted propulsion systems for green shipping: A techno-economic review,” **Ocean**, vol. 2, 2026. 🔗 DOI: 10.26599/OCEAN.2026.9470019.
- 11** R. Haider, W. Shi, Z. Lin, Q. Xiao, **W. Z. Zhang**, and X. Li, “Dynamic response of floating offshore wind turbine to focused wave excitation,” **Physics of Fluids (Q1, IF 3.8)**, vol. 37, no. 8, 2025. 🔗 DOI: <https://doi.org/10.1063/5.0279082>.
- 10** J. Pan, X. Y. Liu, **W. Z. Zhang**, and M. C. Xu, “Numerical study on the dynamical performance of multi-stable metamaterial structures under impact loading,” **Ocean Engineering (Q1, IF 6.3)**, vol. 338, p. 121 938, 2025. 🔗 DOI: <https://doi.org/10.1016/j.oceaneng.2025.121938>.
- 9** M. Song, J. Chen, **W. Z. Zhang***, J. Calderon-Sanchez, and Z. Y. Jiang, “Crashworthiness study of a monopile OWT with metamaterial-based fenders under vessel collisions,” **Engineering Structures (Q1, IF 7.6)**, vol. 345, p. 121 559, 2025. 🔗 DOI: <https://doi.org/10.1016/j.engstruct.2025.121559>.
- 8** J. Tao, W. Chai, X. Yang, **W. Z. Zhang***, C. Wang, and J. Qi, “Stability evaluation of a damaged ship with ice accumulation in arctic regions,” **Journal of Marine Science and Engineering (Q2, IF 2.8)**, vol. 13, no. 9, p. 1685, 2025. 🔗 DOI: <https://doi.org/10.3390/jmse13091685>.

- 7 **W. Z. Zhang**, J. Pan, H. Peng, and M. C. Xu, “Reliability assessment of aged reinforced concrete bridge under ship impact considering multi-hazards,” **Ocean Engineering (Q1, IF 6.3)**, vol. 326, p. 120 772, 2025. [DOI: https://doi.org/10.1016/j.oceaneng.2025.120772](https://doi.org/10.1016/j.oceaneng.2025.120772).
- 6 **W. Z. Zhang***, J. Calderon-Sanchez, T. Almeida-Medina, A. Medina-Manuel, and A. Souto-Iglesias, “Numerical analysis of forced heave oscillations for an FOWT column with a heave plate using overset mesh,” **Ocean Engineering (Q1, IF 6.3)**, vol. 341, p. 122 736, 2025. [DOI: https://doi.org/10.1016/j.oceaneng.2025.122736](https://doi.org/10.1016/j.oceaneng.2025.122736).
- 5 **W. Z. Zhang**, J. Pan, J. Calderon-Sanchez, X. B. Li, and M. C. Xu, “Review on the protective technologies of bridge against vessel collision,” **Thin-Walled Structures (Q1, IF 8.3)**, vol. 201, p. 112 013, 2024. [DOI: https://doi.org/10.1016/j.tws.2024.112013](https://doi.org/10.1016/j.tws.2024.112013).
- 4 **W. Z. Zhang***, J. Calderon-Sanchez, D. Duque, and A. Souto-Iglesias, “Computational Fluid Dynamics (CFD) applications in Floating Offshore Wind Turbine (FOWT) dynamics: A review,” **Applied Ocean Research (Q1, IF 5.1, Highly Cited Paper)**, vol. 150, p. 104 075, 2024. [DOI: https://doi.org/10.1016/j.apor.2024.104075](https://doi.org/10.1016/j.apor.2024.104075).
- 3 J. Pan, **W. Z. Zhang**, Z. M. Sun, X. Qu, and M. C. Xu, “Experimental study on the dynamical response of elastic trimaran model under slamming load,” **Journal of Marine Science and Technology (Q1, IF 2.7)**, pp. 1–16, 2023. [DOI: https://doi.org/10.1007/s00773-023-00969-y](https://doi.org/10.1007/s00773-023-00969-y).
- 2 J. Pan, T. Wang, **W. Z. Zhang**, S. W. Huang, and M. C. Xu, “Study on the assessment of axial crushing force of bulbous bow for bridge against ship collision,” **Ocean Engineering (Q1, IF 6.3)**, vol. 255, p. 111 411, 2022. [DOI: https://doi.org/10.1016/j.oceaneng.2022.111411](https://doi.org/10.1016/j.oceaneng.2022.111411).
- 1 M. C. Xu, Z. J. Song, T. Wang, **W. Z. Zhang**, and J. Pan, “Empirical formulae assessment of ultimate strength for perforated stiffened panels under longitudinal compression,” **Ocean Engineering (Q1, IF 6.3)**, vol. 264, p. 112 445, 2022. [DOI: https://doi.org/10.1016/j.oceaneng.2022.112445](https://doi.org/10.1016/j.oceaneng.2022.112445).

Conference Proceedings

- 4 **W. Z. Zhang***, J. Calderon-Sanchez, and A. Souto-Iglesias, “Influence of diameters in double heave plates on hydrodynamic coefficients under constant material usage,” in *International Conference on Offshore Mechanics and Arctic Engineering (OMAE2026)*, American Society of Mechanical Engineers, vol. 180583, 2026.
- 3 C. Wang, J. Pan, **W. Z. Zhang**, and M. Xu, “Study on dynamical response of improved cylindrical shell chain structure under impact load,” in *Trends in Collision and Grounding of Ships and Offshore Structures*, CRC Press, 2025, p. 317.
- 2 J. Pan, H. Peng, J. Gan, **W. Z. Zhang**, and X. Li, “Study on crashworthiness performance of an airbag anti-collision device for vessel-bridge collision,” in *International Conference on Offshore Mechanics and Arctic Engineering (OMAE2023)*, American Society of Mechanical Engineers, vol. 86847, 2023, V002T02A004.
- 1 **W. Z. Zhang**, J. Pan, N. Li, and M. Xu, “The safety assessment of ship-bridge collision based on a simplified dynamic model,” in *International Conference on Offshore Mechanics and Arctic Engineering (OMAE2023)*, American Society of Mechanical Engineers, vol. 86847, 2023, V002T02A002.

Academic roles

2024-Now	Journal peer reviewer Over 40 reviews for top journals in the field of ocean engineering and offshore renewable energy. <i>Applied Energy, Marine Structures, Ocean Engineering, Engineering Conversion and Management, Applied Ocean Research, Journal of OMAE.....</i>
2025-2027	Master thesis reviewer & supervisor & committee & lecturer <i>EMSHIP</i> , International Master program in Advanced Design of Sustainable Ships and Offshore Structures. <i>MAERM</i> , Master’s Degree on Marine Renewable Energies Harnessing.

Selected Awards

- 2026 **National Award**, Chinese Government Award for Outstanding Self-Financed Students Abroad.
45th OMAE Outreach Scholarship, Outreach program and financial aid by the 45th OMAE Conference. (Only 11 doctoral recipients from a global pool of 900+ applicants)
- 2025 **National Scholarship**, Financial Aid for Predoctoral Research Staff 2025 supported by Santander Scholarship.
- 2024 **National Thesis Award**, Outstanding Thesis in Hydraulic Engineering Majors of National Higher Education Institutions in China.
- 2023 **National Scholarship**, Spanish government FPI Scholarship for pre-doctor researcher.
Excellent Thesis Award, Top 1% postgraduate thesis in Wuhan University of Technology.
Excellent Postgraduate Award, Top 1% postgraduate in Wuhan University of Technology.
- 2020 **Excellent Bachelor Award**, Top 1% bachelor in Wuhan University of Technology.
- 2019 **National 1st Level Award**, 8th China & International Marine Vehicle Design and Construction Contest.
- 2018 **National 2nd Level Award**, 7th China & International Marine Vehicle Design and Construction Contest.

Funding and projects

- 2025-2028 **Spanish National Plan Funding: RENEWFLOAT**, Budget: €125,000
Lead researcher, Design of shared mooring and multi-use floating renewable platforms, funded by the Ministry of Science, Innovation and Universities (MCIU) of Spain, PID2024-161775OB-C21.
- 2026-2027 **Shanghai Sci-tech Co-research Program: China-Spain collaboration**, Budget: ¥400,000
Lead researcher from Spanish side, Study on the Coupling Mechanism Between Fishing Gear Mesh Structure and Catch Escape Size Based on a Hydrodynamic Coupling Model and the Escape Threshold. Funded by Shanghai Municipal Government, 25HB2710100.
- 2023-2025 **Spanish National Plan Funding: UPM-FOWT-DAMP2**, Budget: €121,000
Lead researcher, Hydrodynamics of Motion Damping Devices for Floating Offshore Wind Turbines: Numerical modeling and small-scale experiments, funded by the Ministry of Science, Innovation and Universities (MCIU) of Spain, PID2021-123437OB-C21.
- 2024-2028 **European COST Project: Decarbonising Waterborne Transportation (DeWaTra)**
Participant, European Cooperation in Science and Technology (COST grant), CA23159, DeWaTra WG4: Renewable energy contributions to ship energy efficiency.

Skills

- Languages **Mandarin Chinese**-Mother language; **English**-Fluent; **Spanish**-Basic
- Computer skills **OpenFOAM** (CFD simulations of fluid-structure interaction, waves, FOWTs), **MATLAB** (Post-processing of data), **ANSYS LS-DYNA** (FEA simulations of impact and collision), **Solidworks & Rhino** (Construction of solid models), **LaTeX**, ...